

Lab 1

Briefing

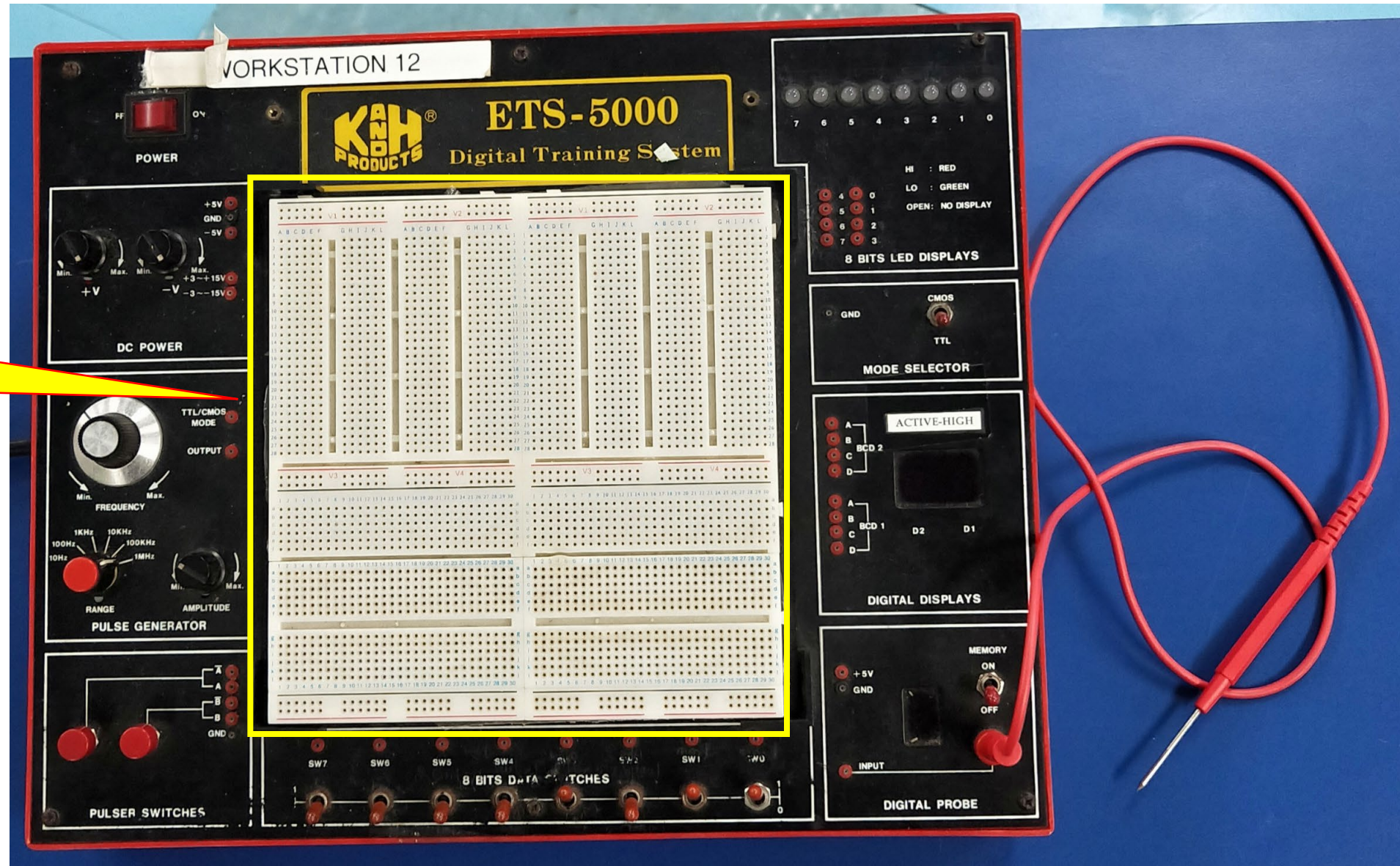
ETS-5000

Power
button

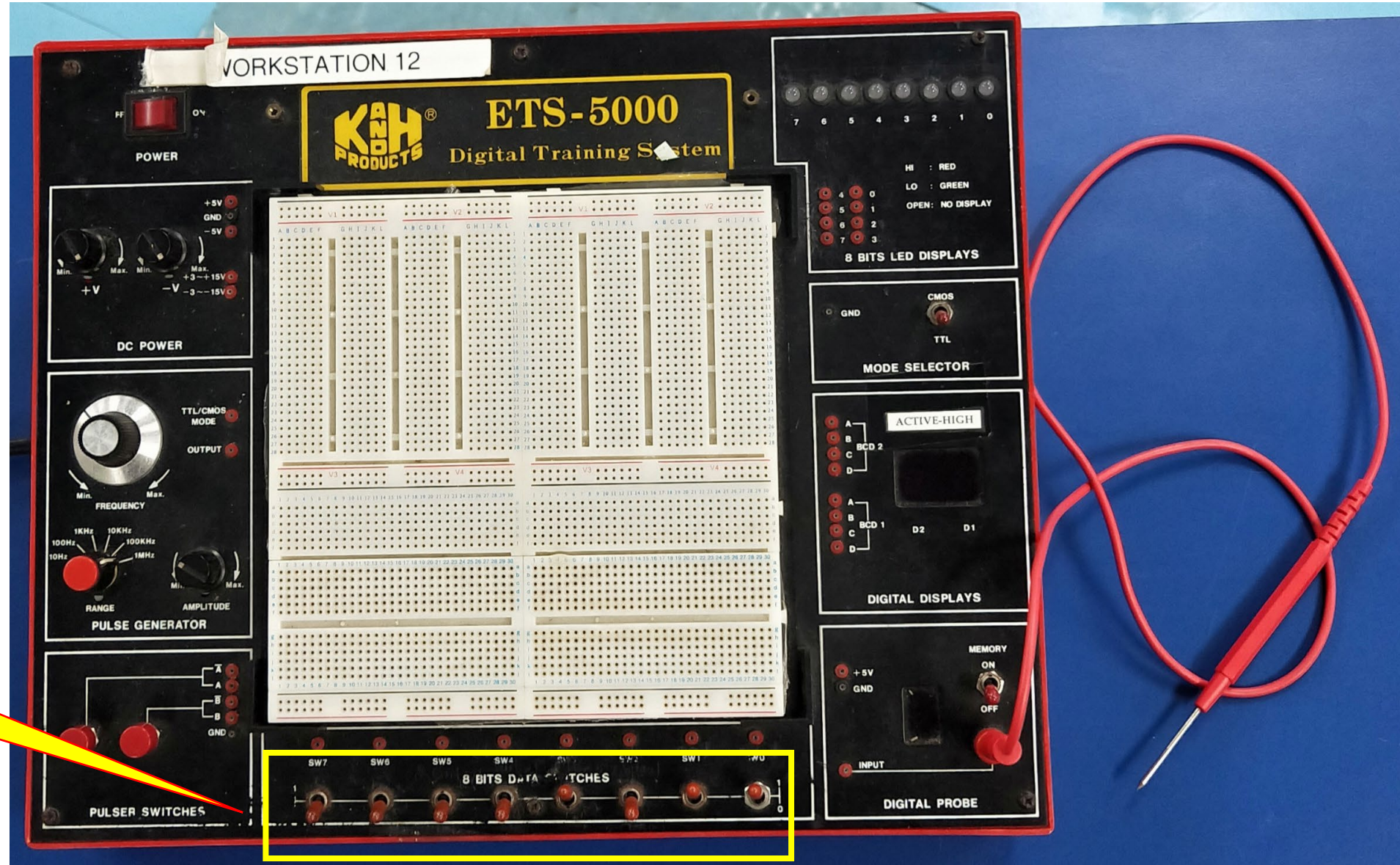


ETS-5000

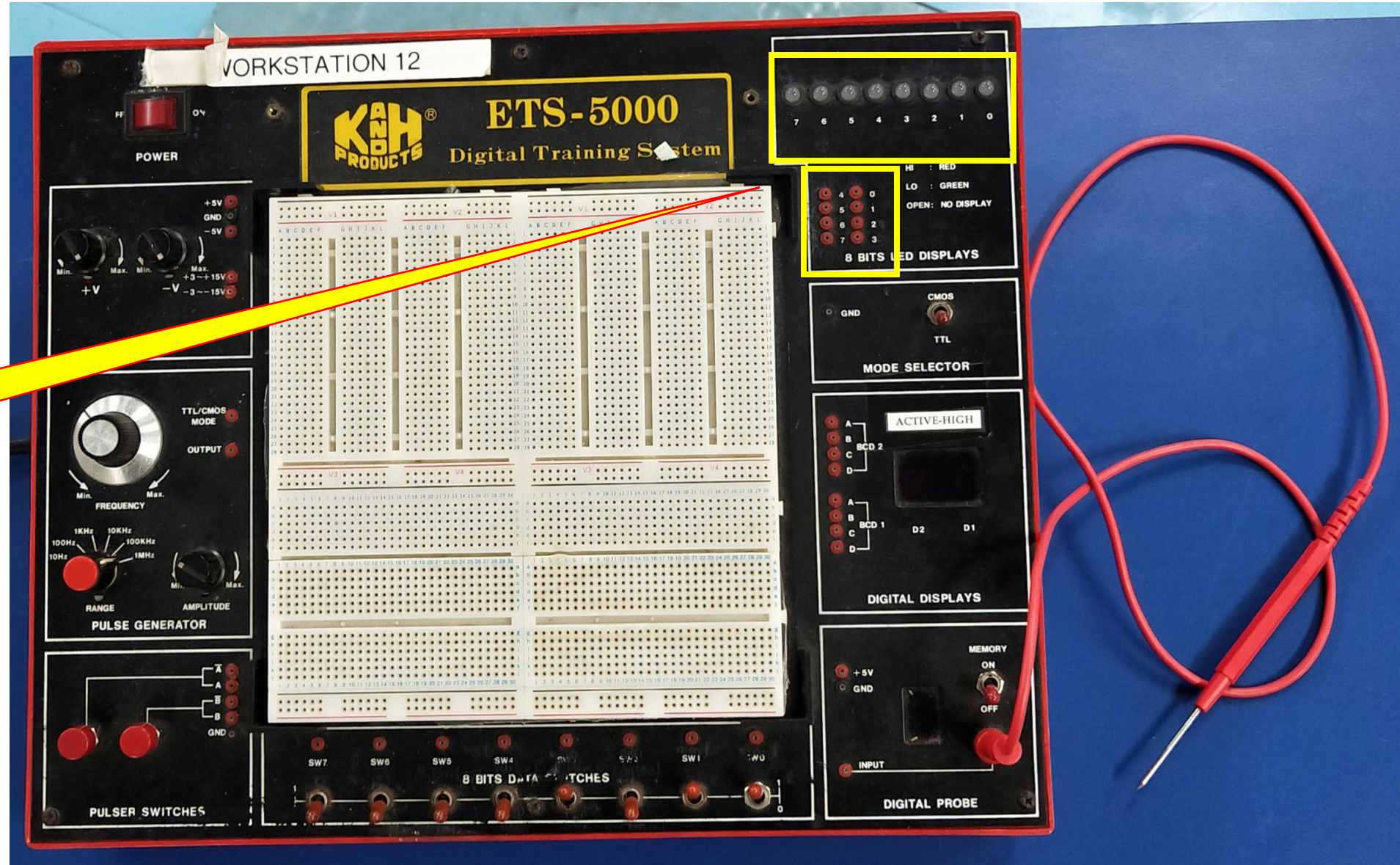
Breadboard



ETS-5000

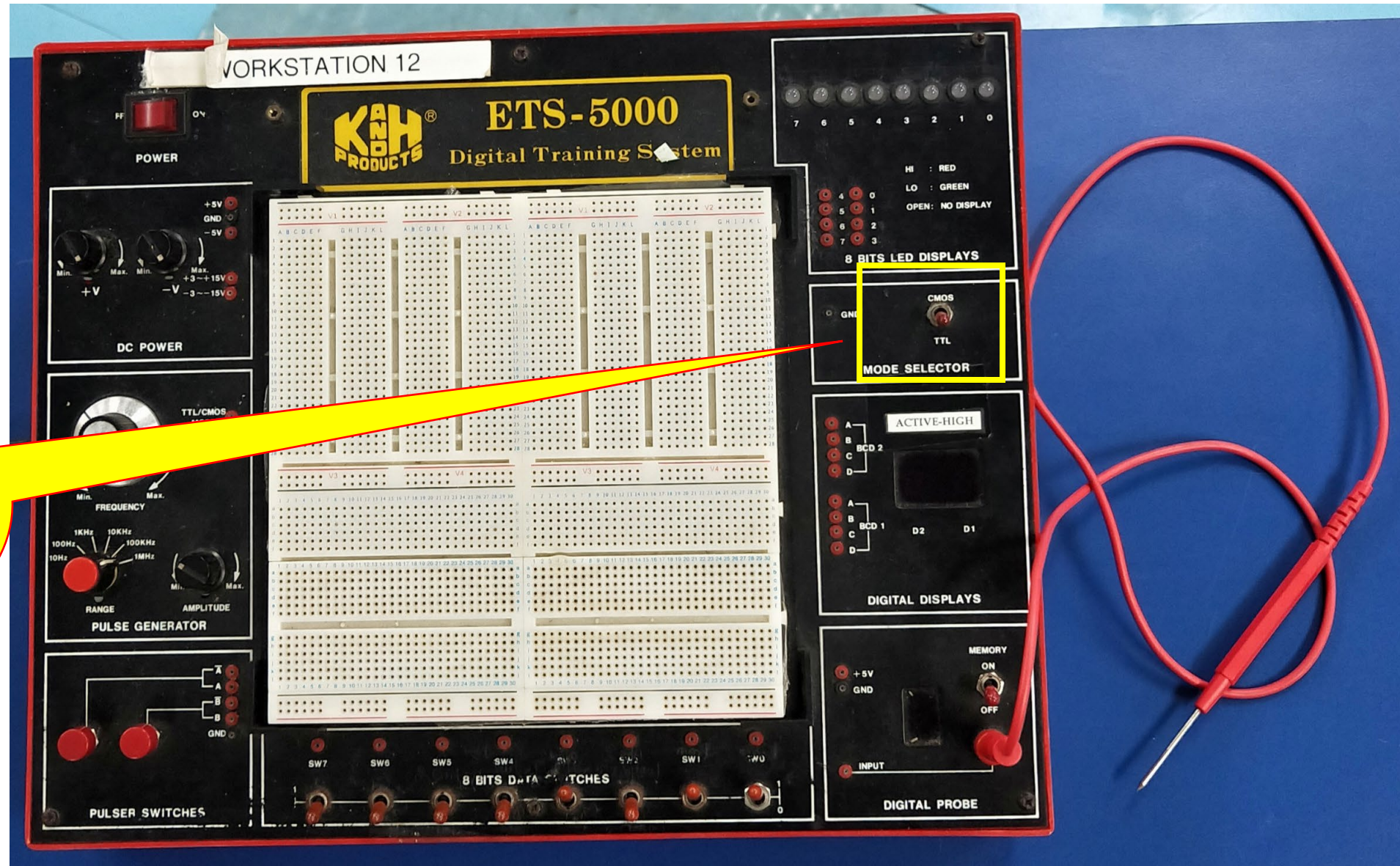


ETS-5000



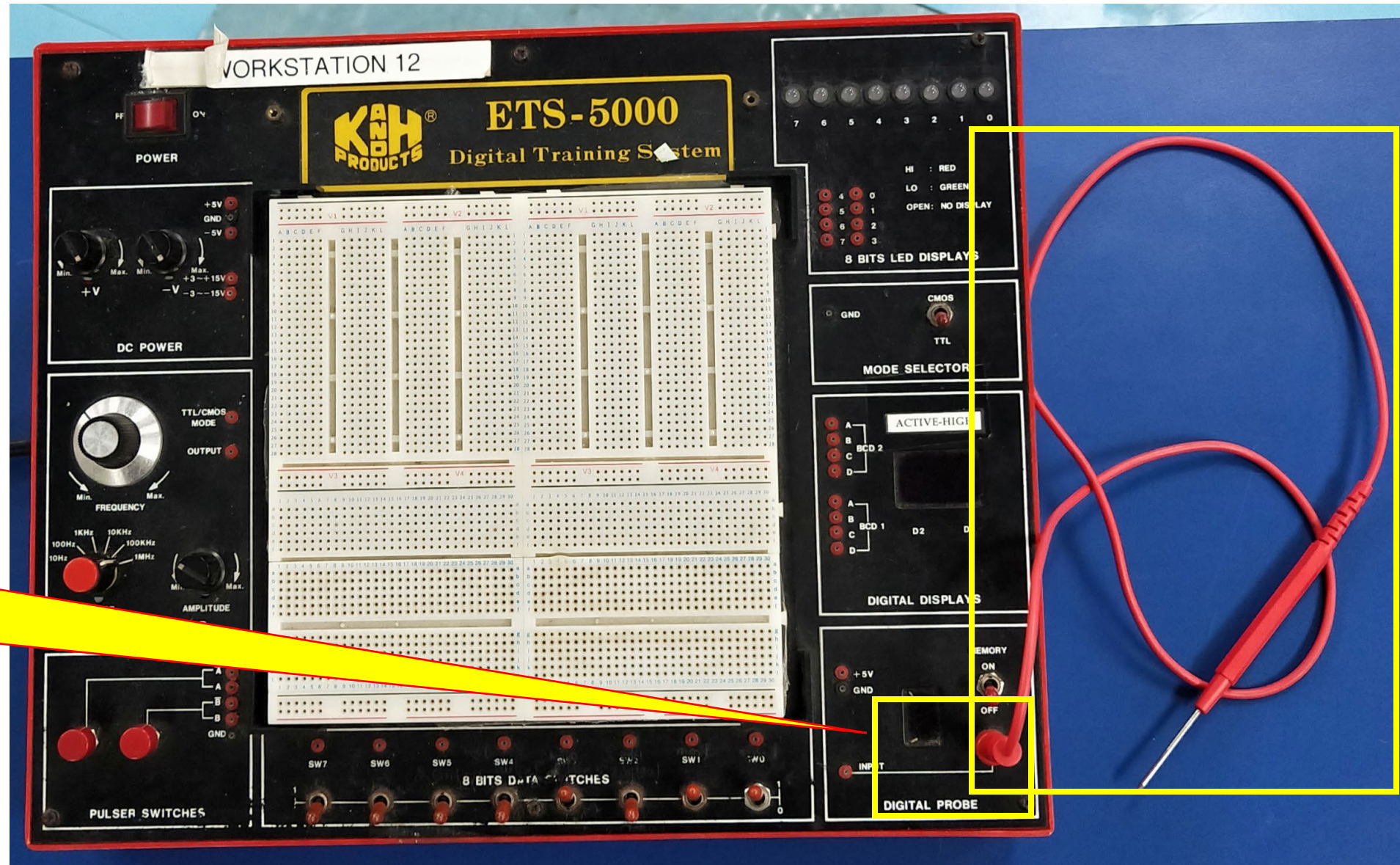
Output
LEDs

ETS-5000



ETS-5000

Digital Probe



Breadboard Layout

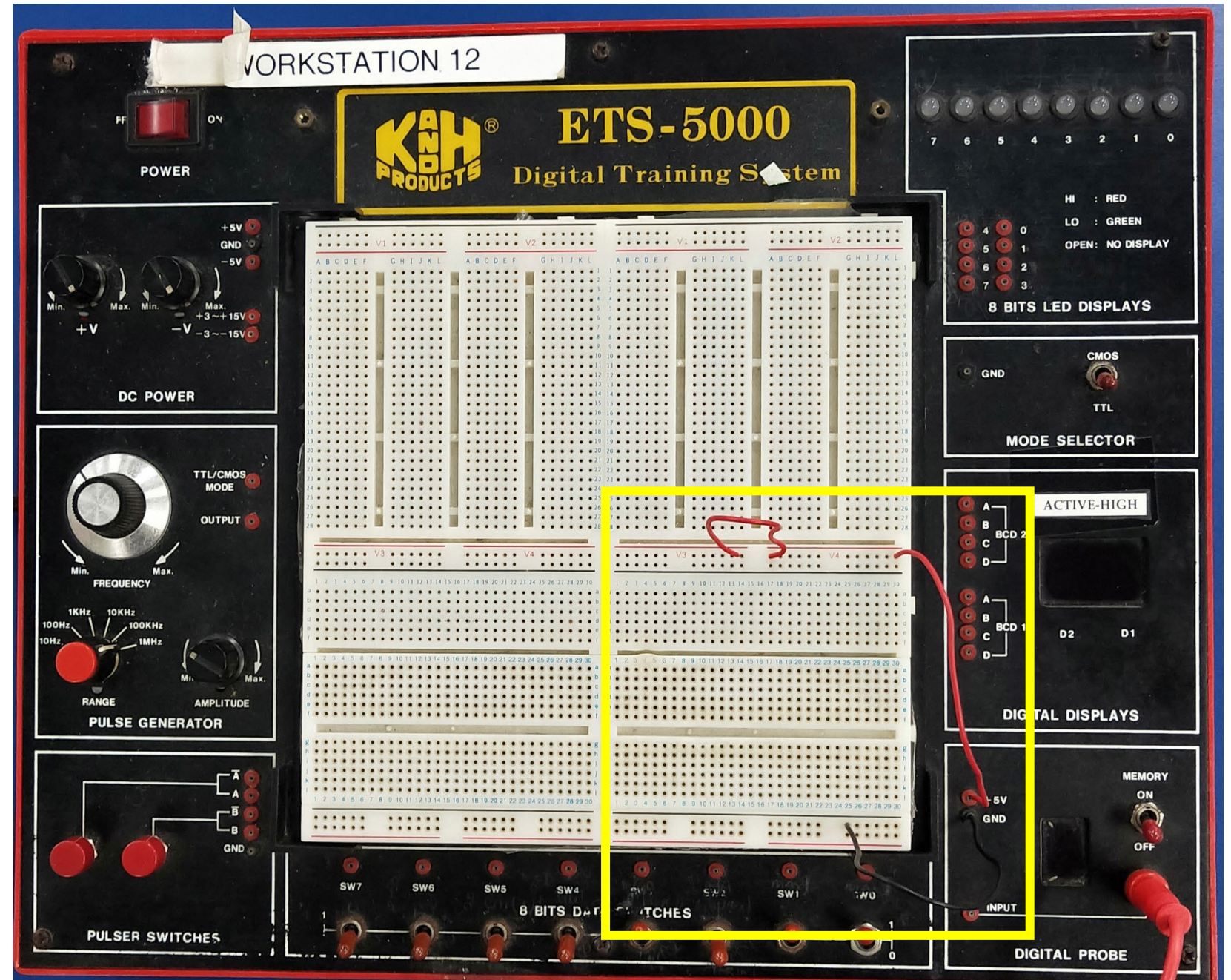
Each rows / columns of holes are electrically connected internally

Connect to
Vcc

Connect to
Gnd

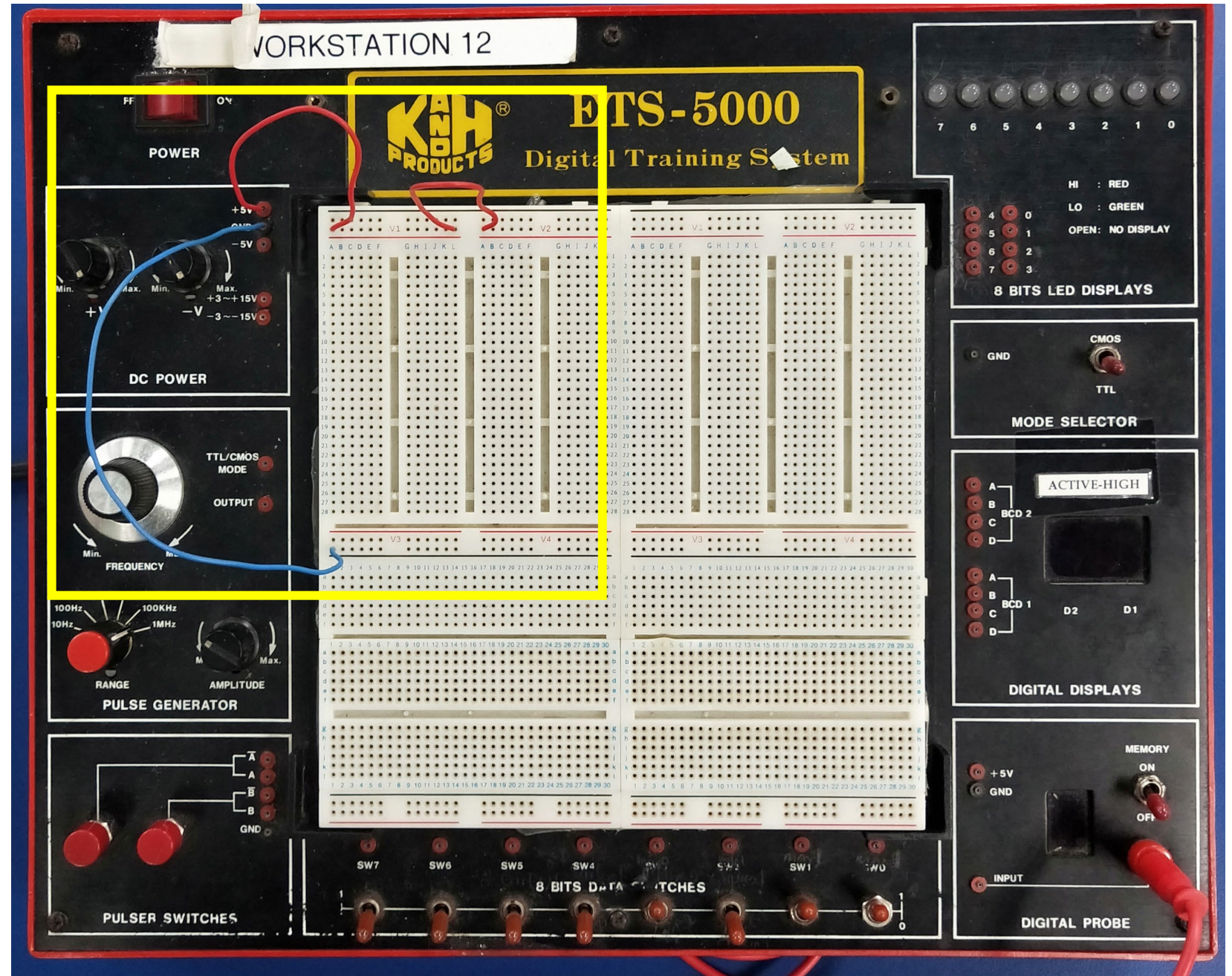


Option #1



Vcc and GND
wires
connection on
ETS-5000

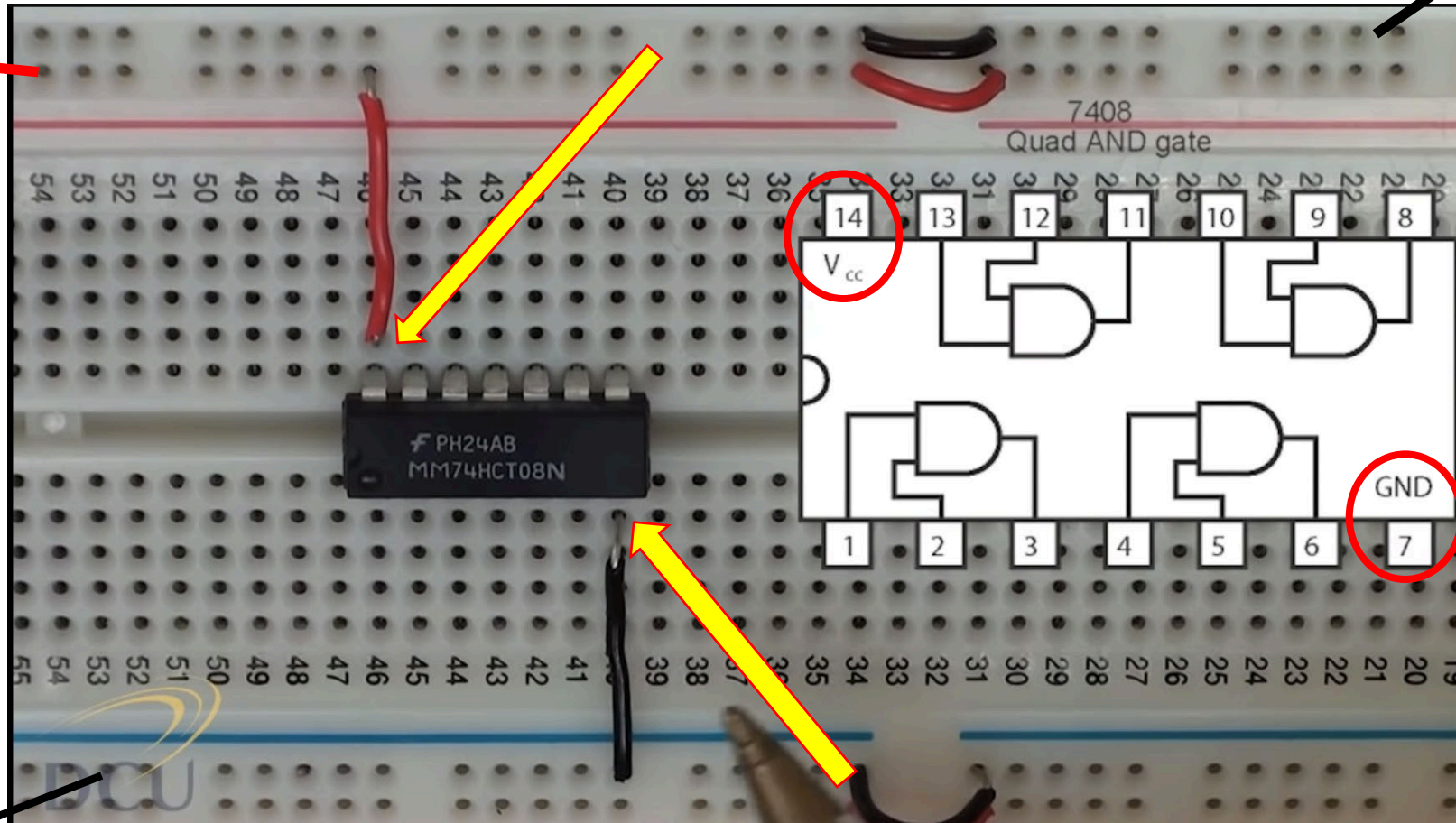
Option #2



Part 1: Pins Layout: Vcc – pin 14, Gnd – pin 7

Connect
to Vcc

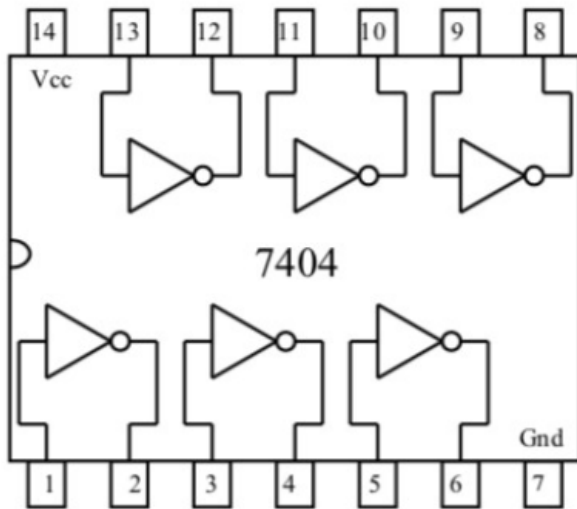
Connect
to Gnd



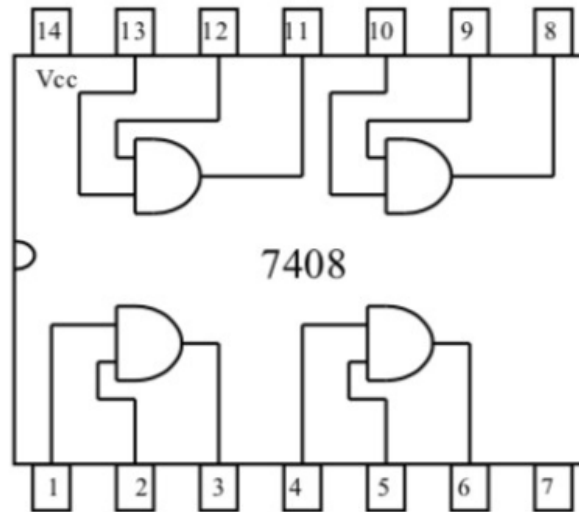
7408 IC
(AND gate)

Connect
to Gnd

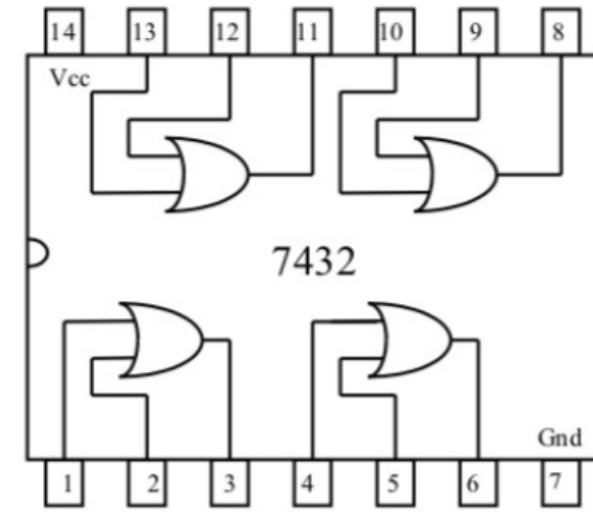
Pins Layout: AND, INVERTER and OR gates ICs



7404 Hex Inverter

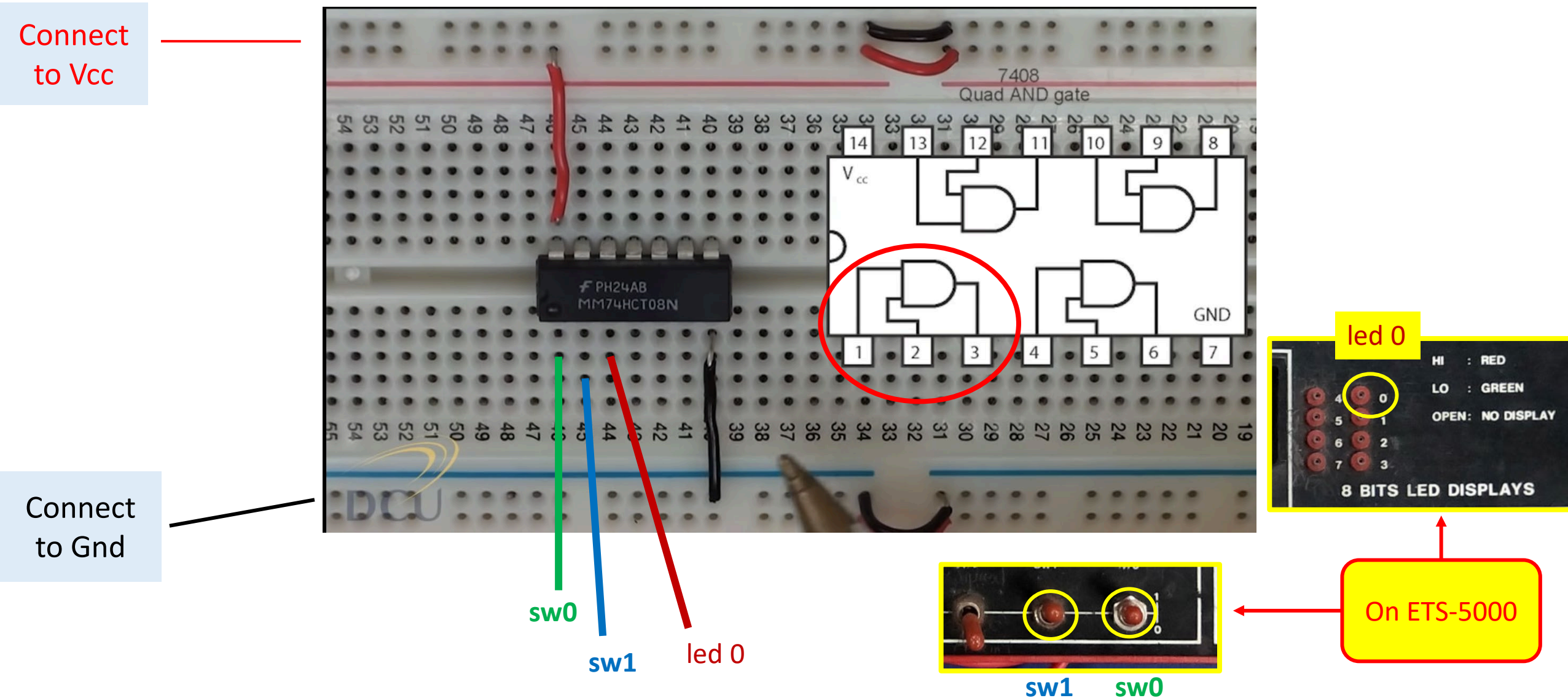


7408 Quad 2-input AND

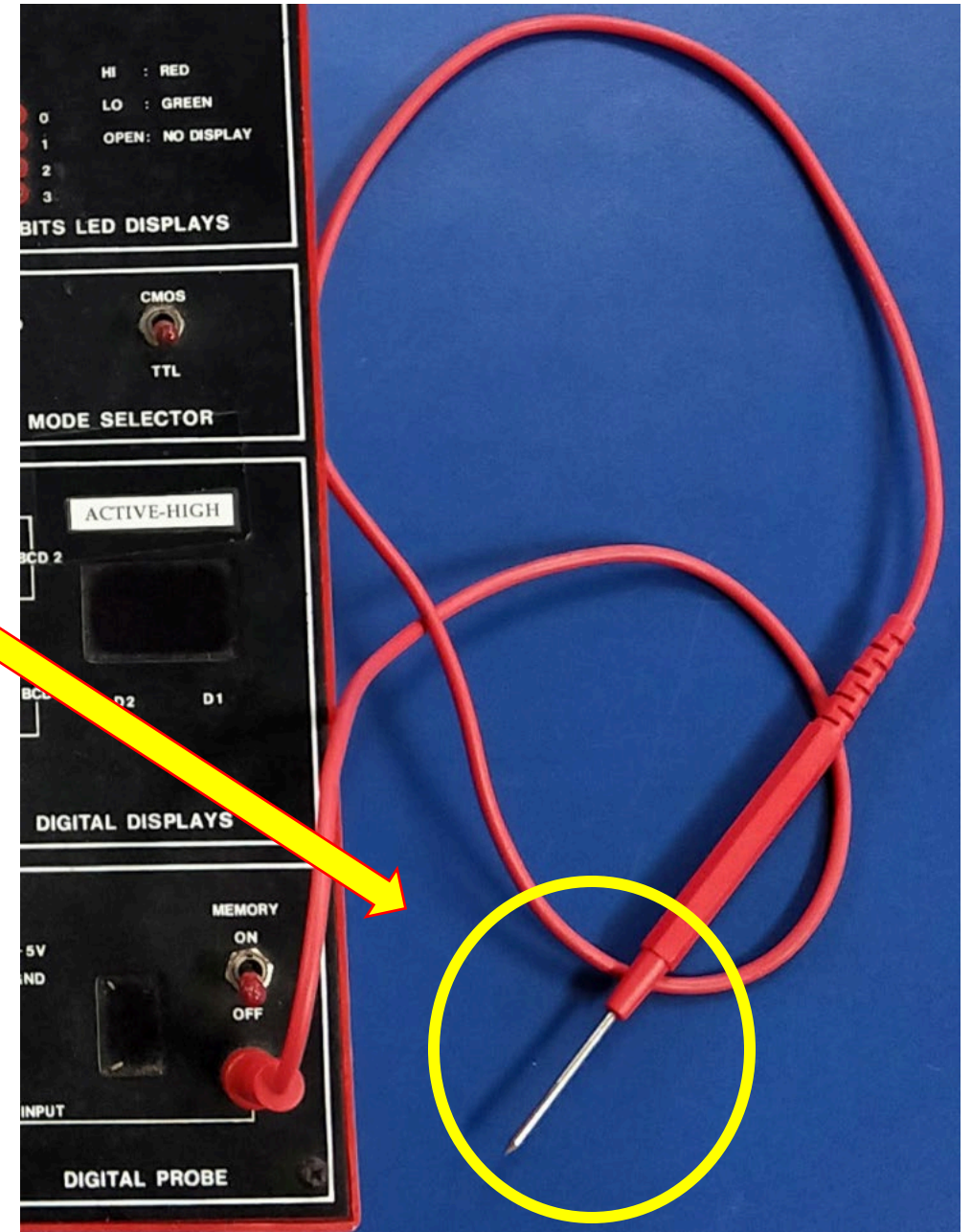


7432 Quad 2-input OR

Pins Layout: sw1 – pin 1, sw2 – pin 2, led 1 – pin 3

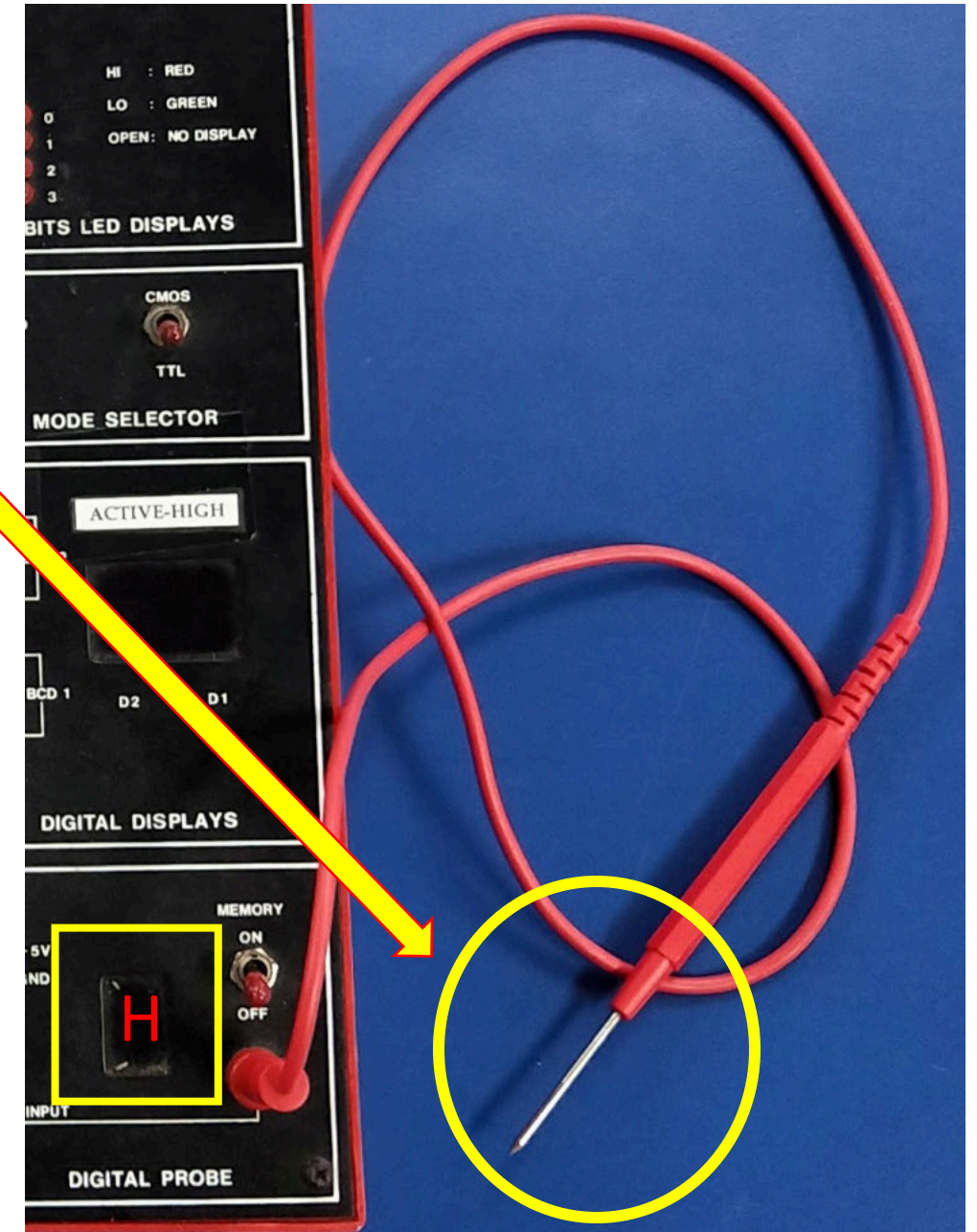
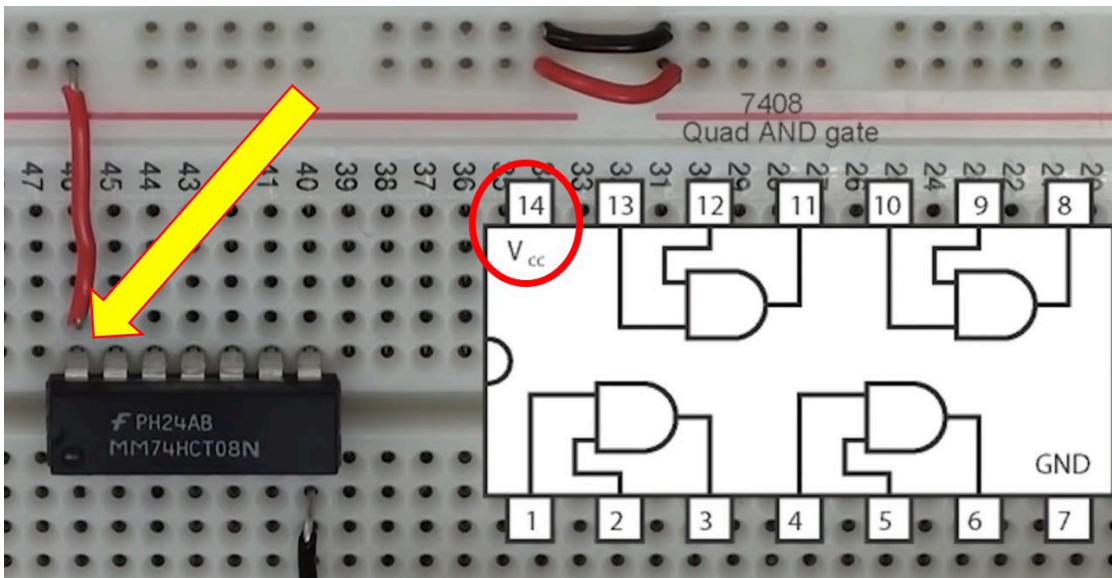


Use 'Digital Probe tip' to
remove ICs on the
breadboard!



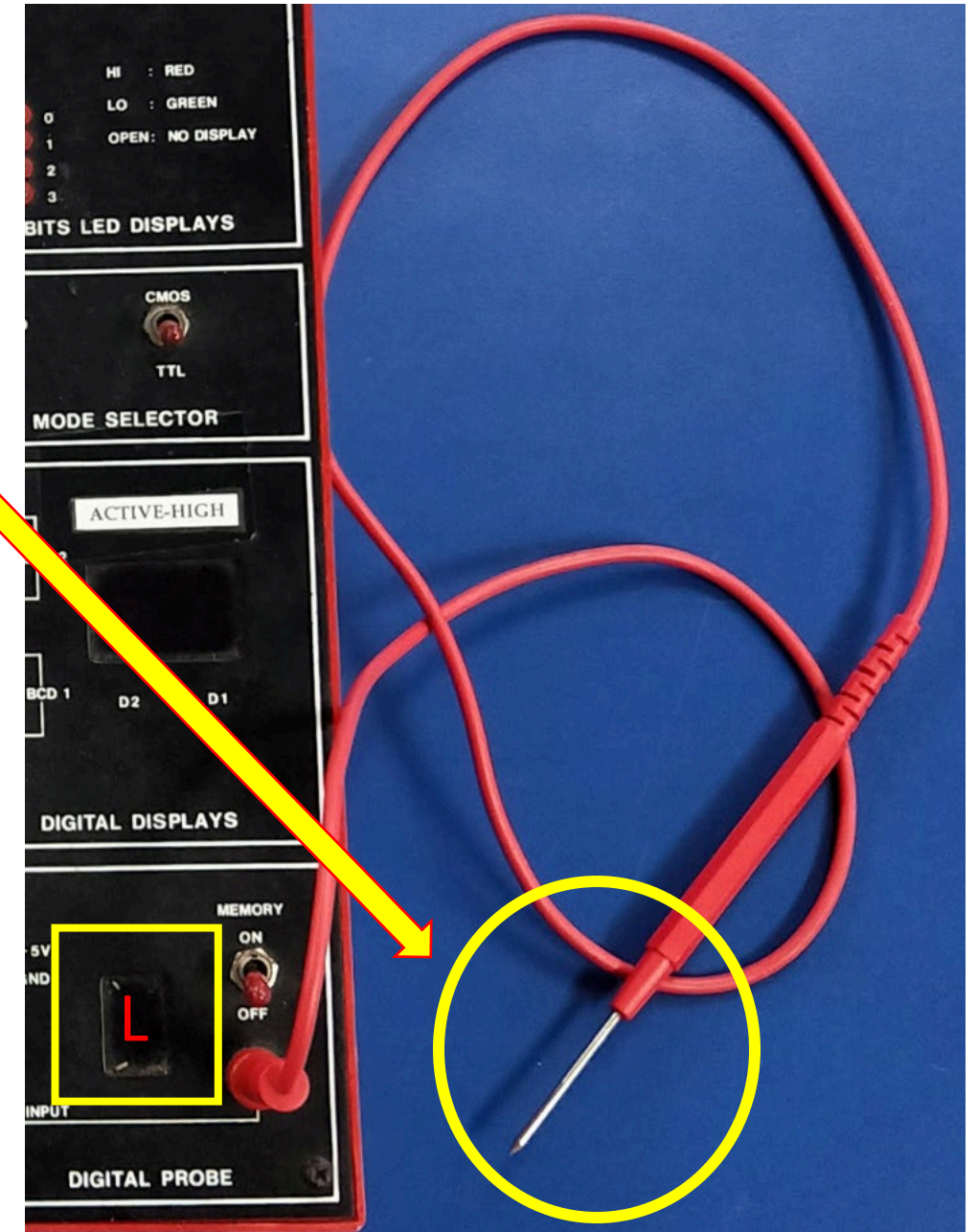
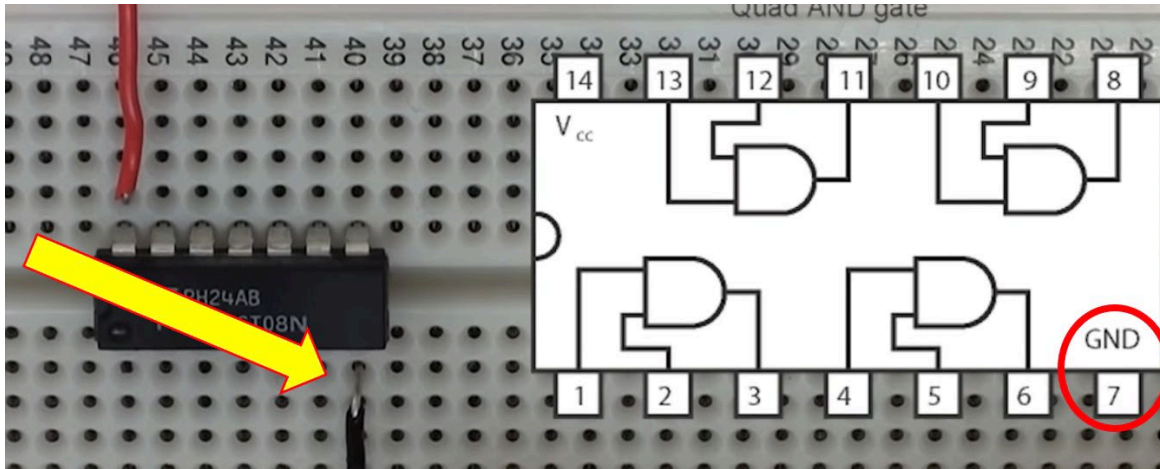
Use 'Digital Probe tip' to diagnose ICs connection on the breadboard:

- touch the tip on IC pin no 14 (V_{cc}), Digital Probe **led** will display **H** that indicate high voltage, only if the connection is good / correct

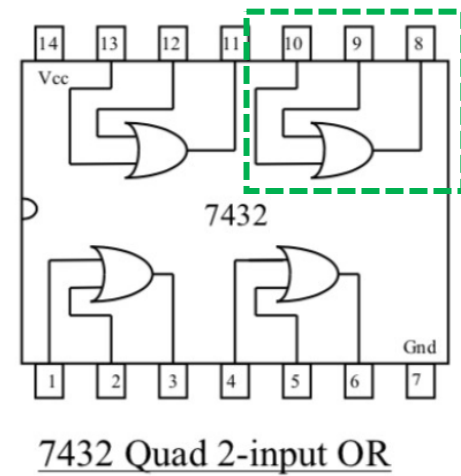
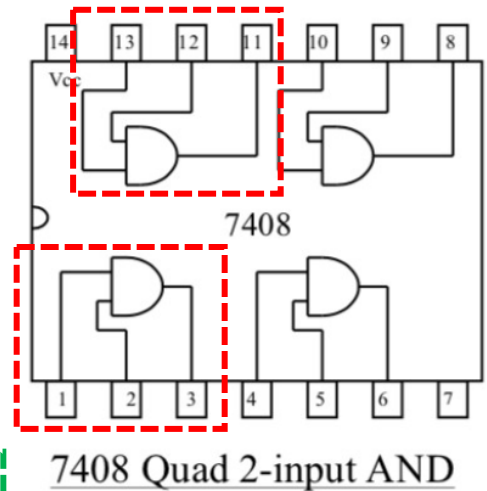
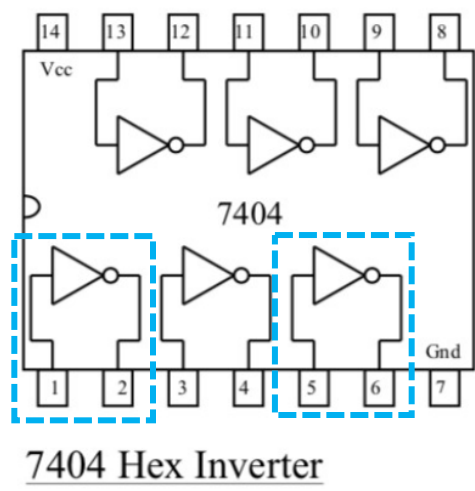
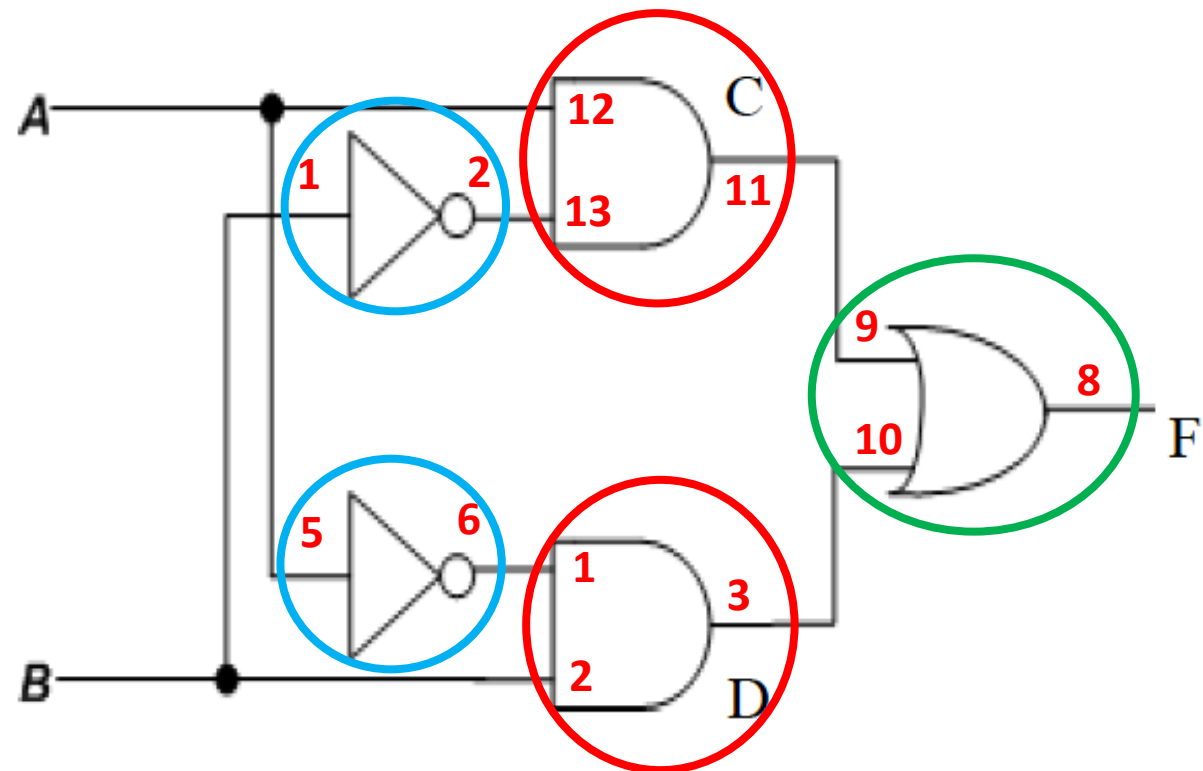


Use 'Digital Probe tip' to diagnose ICs connection on the breadboard:

- touch the tip on IC pin no 7 (GND), Digital Probe **led** will display **L** that indicate low voltage, only if the connection is good / correct



Part 3: Label your circuit diagram with pin numbers.
(The labels shown on the circuit diagram below are for reference/suggestion.)



You may start now ...

happy breadboarding ...